

NOOK Tablet 7

Barnes & Noble releases a new Android tablet pre-loaded with NOOK software for just \$50. <http://dgit.in/BnNnuktb>

**AI recognises objects**

A new AI project from Disney Research can now identify various cars, birds and animals. <http://dgit.in/AIdsnrys>

for the discovery has also designed and built non-hygrovoltaic 'evaporation engines' which are mechanical devices. One of them is a wheel to directly generate rotatory motion from the nanoscale harnessing of evaporation. The wheel rotates using a gradient in humidity to keep one side heavier than the other. If this technology can be enhanced and scaled up, it will be a promising source of alternative energy, even among rural or third world communities devoid of affluence.

Bacteria batteries

There are billions of impoverished people in the world, many of them living in remote and undeveloped third world areas. With medical facilities few and far between, life there is often rife with disease. Half the fight against disease is diagnosing it in the first place, and that is also where a majority of the cost of treatment lies with most current diagnostics being done in labs. A whole generation of scientists are working to change that, by working to develop extremely low cost and easily mass producible devices using micro- and nano-fluidics so that diagnosis can be done on site in a jiffy and at almost no cost. One such scientist of this honourable generation is Sean Choi, who is Assistant Professor of Electrical and Computer Engineering at Birmingham University. His work also focuses on the development of microfluidic point of care devices, and apart from biosensors for detection of proteins and nucleic acids he is tackling another crucial aspect of the devices, namely, how to power them. As a result of his research (along with his colleagues), we have paper based microbial fuel cells, or in colloquial parlance, a paper bio-battery. The invention is a piece of paper printed with a carbon based cathode on one end and a nickel based anode on the other, which is almost ready to produce electricity when folded up. The first and final requirement to get the battery started is a few drops of dirty pond water! While the output isn't nearly enough to charge your smartphone, there are potatoes

for that, so stop thinking of your first world problems and imagine how many third world lives could be saved. The bio-battery works thanks to the bacteria in the water, utilising microbial respiration to generate a few microwatts of power, which is sufficient for the purpose of handling the electronics of microfluidic devices. The bacteria batteries could ultimately lead to a self-sufficient disposable diagnostic device for common diseases, helping trained doctors evaluate ailments at rates never seen before.

Bio-concrete

Concrete is mankind's favourite building material, and we have been using it to erect everlasting structures and pave roads ever since the Romans, more than two millennia ago. Unfortunately, the material isn't really that everlasting, no matter what special



Soon such cracks will heal automatically and disappear.

ingredients or mixing techniques are used. The world is a harsh environment with widely varying conditions over time, and not unlike a rubber band even concrete eventually cracks. To make matters worse, water seeps into the said cracks and begins to corrode the steel bars of the building's skeleton, eventually leading to spontaneous

collapse (not the 9/11 kind). For all our structural reinforcements and additives, we haven't solved the problem at its root. Until now, that is. Okay, well to be honest, it has been a few years. As the story goes, back in 2006, Eric Schlangen who is a concrete technologist, asked Henk Jonkers, a microbiologist and this paragraph's protagonist, if it would be possible to make self healing concrete. More than three years and a lot of thought and experiments later, Jonkers had successfully devised a solution while working at Delft University of Technology in the Netherlands. His solution for repairing cracked concrete, of course, utilised the only thing in nature that repairs itself - Life. The problem at first was finding a life form that could survive and thrive inside concrete, an environment which is quite dry and highly alkaline. It also had to easily fit inside the tiniest of cracks, so the obvious life form of choice was bacteria. Moreover, life forms are made of tissue hence produce tissue to regenerate, so the chosen bacteria had to produce something more structurally useful that could repair concrete. Jonkers also had to give them something to feed off of, in order to multiply, but the food had to be something that didn't make the concrete weak. After mixing various bacteria in concrete, they found three strains that survived. Ultimately *Bacillus* spores were chosen (aren't they some multitasked microbes!), as they lay perfectly dormant till exposed to rainwater that seeped in through the cracks. The food of choice was calcium lactate, which the bacteria consumed after germinating, producing calcium carbonate (limestone) in the process. The spores and calcium lactate are stored in a biodegradable casing that is mixed into the concrete. There they lay waiting to see the light of day, only to block it out as soon as they do so!

Reproductive hard drives

Genetic memory has already been manufactured by scientists at leading research institutes around the world, who stored information in synthesised